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Macroeconomic impact of FDI inflows: an ARDL approach for the case of India

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ABSTRACT

This article investigates the dynamics of relationship between foreign direct investment (FDI) inflows and major macroeconomic variables, i.e. gross domestic product (GDP), exports and exchange rate for India. Using annual time series data, the empirical analysis has been carried out for the period 1980–2012. Using the most recent autoregressive distributed lag (ARDL) testing approach to cointegration proposed by Pesaran et al., 2001, the study concludes that there is a strong evidence of long-run relationship between variables with GDP, FDI inflows and exports as the dependent variable. However, there is lack of evidence of long-run cointegration with exchange rate as a dependent variable. The results indicate bilateral positive and significant relationship between FDI inflows and GDP; and GDP and exports in the long run and short run. In contrast, the result analysis indicates negative impact of FDI inflows on exports in the long run. Moreover, exchange rate appreciation has a positive impact on FDI inflows and exports, but negative on GDP in the long run. The error-correction term (ECT) for the models signifies a fairly quick speed of adjustment to the equilibrium following a shock. Furthermore, the study also observes long-run causality running from the explanatory variables towards the respective dependent variables.

KEYWORDS

Foreign direct investment; gross domestic product; exports; exchange rate; India

1. Introduction

Foreign direct investment (FDI) has been assumed an increasingly important role in international business in the wake of deepening financial and economic integration between developing and developed countries. This in turn is the result of several factors, such as expansion of information technology, financial liberalisation, innovation and spread of institutional investors. The FDI flows are a preferred form of foreign investment for developing countries compared to foreign portfolio investment as they are believed to have a positive impact on a number of macroeconomic variables such as gross domestic product (GDP) and exports. At the same time, a surge or reversal in the FDI flows can complicate macroeconomic management or basic fundamentals of an economy. This may also develop financial and economic risks for the host country. A surge may lead to an appreciation/depreciation of the exchange rate and/or undermine the competitiveness of the domestic tradable sector, possibly causing long lasting damage (with significant adverse effects in case of a sudden reversal).

Studies relating to developing countries shows that the FDI plays an imperative role in long-term economic development of the country by augmenting the supply of capital, strengthening infrastructure, transferring technology, generating new employment opportunities thereby enhancing the competitiveness of the domestic economy and boosting productivity (Alfaro et al., 2004; Balasubramanyam et al., 1996; Basu et al., 2003; Bhasin, 2012; Blomstrom et al., 1994; Borensztein et al., 1998; Chowdhury & Mavrotas, 2005; Hansen & Rand, 2006; Jajri, 2009; Kumar & Pradhan, 2002; Makki & Somwaru, 2004; Nair-Reichert & Weinhold, 2001; Zhang, 2001). Meanwhile, it throws the risk of destroying local capabilities and/or exploiting host countries' resources or may act as a neutral one (Aitken & Harrison, 1999; Carkovic & Levine, 2002; De Mello, 1999; De Mello & Fukasaku, 2000; Haddad & Harrison, 1993; Johnson, 2006).

India has emerged as one of the most preferred FDI destination by transnational corporations (TNCs) in recent times. The *World Investment Prospects* (WIP) Survey 2013–2015 (*UNCTAD*, 2013) ranked India at number three among the TNCs' top prospective host economies next only to China and the United States. India has witnessed a

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high growth and development with remarkable buoyancy since 1981. From this time, initial economic reforms were initiated, followed by phased and progressive opening of the economy to international investment and trade in subsequent years. The FDI inflows to India have been steadily growing since the last two to three decades. In the wake of the perceived importance of the FDI inflows for a developing economy like India, it becomes more imperative to examine if the rising FDI inflows are actually having the desired impact on important macroeconomic variables like GDP, exports and exchange rate. Through this article, we examine the relationship between the FDI inflows in India and major macroeconomic variables, such as GDP, export and exchange rate for the period 1981–2012, using the robust autoregressive distributed lag (ARDL) approach. In particular, the main objectives of the study are:

i. to examine whether there exists a long-term relationship between the FDI inflows and select macroeconomic variables, i.e. GDP, exports and exchange rate, and
ii. to identify the causal relationship among the FDI inflows and GDP, exports and exchange rate in India.

2. Literature review

There is considerable literature examining the relationship of the FDI inflows as well as outflows with individual variables like GDP or exports (Bhasin & Paul, 2016; Carkovic & Levine, 2002; Chowdhury & Mavrotas, 2005; Hansen & Rand, 2006; Johnson, 2006; Kumar & Pradhan, 2002; Nair-Reichert & Weinhold, 2001). Most of the studies (theoretical and empirical findings) in the past examined the role of the FDI inflows as a growth-enhancing factor for the recipient economy. Campos and Kinoshita (2002) observe that 'The positive impact of the FDI on economic growth seems to acquire status of stylized fact in the international economics literature'. There is a fair amount of literature examining individually the relationship between the FDI inflows and growth/GDP/export led growth (ELG), as well as growth and exports linkages using different research techniques. The literature reviews has been divided into different sub-sections, i.e. studies relating to both developed and developing countries; studies relating to developing countries and studies relating to India. Further, we look at studies examining the relationship between the FDI inflows and/or GDP and/or exports; and **s**tudies examining the relationship between the FDI inflows and exchange rate. A summarised review of the literature has been presented in Table 1. The review of literature also shows that relatively sparse work has been done in examining the dynamics of relation of exchange rate with other macro-economic variables or the FDI inflows.

3. Research methodology

This study has incorporated the most important variables, which may be affected by the inflows of FDI and/or acts as decisive factors for foreign investors to invest in the Indian markets. Therefore, the study examines the nature of relationship (long-run and short-run) that exists between the FDI inflows, GDP, exports and exchange rate during the period 1981–2012. The data for the FDI inflows has taken from the *UNCTAD Database* (2013); United Nation Conference on Trade and Development. The data for the GDP, exports and exchange rate has taken from the *Handbook of Statistics on the Indian Economy, 2013*, Reserve Bank of India.

The study employs the more recent and robust ARDL bound testing approach suggested by Pesaran et al. (2001) to study the long- and short-run relationships among these four time series variables. The conventional cointegration method, such as Johansen (1988) and Johansen and Juselius (1990) approaches, estimates the long run relationships within the context of a system of equations. The ARDL method employs a single reduced form of equation. Further, this approach does not involve pre-testing of variables, which means that it is immaterial whether the underlying regressors are purely I(0), purely I(1) or a mixture of both. One must be careful about the fact that in ARDL, the variables should not be of I(2), otherwise it may lead to spurious results.

In addition, the ARDL method avoids the large number of specifications to be made, such as the number of endogenous and exogenous variables (if any) to be included, the treatment of deterministic elements, as well as the order of VAR and the optimal number of lags to be used. More importantly, the ARDL technique has known to have small sample size properties in presenting more robust and reliable results. This model can be used with limited or small sample size data (i.e. 30–80 observations), with a set of the critical values originally given by Narayan (2005). In addition, the long-run and short-run parameters of the model are also estimated simultaneously.

Table 1. Studies examining the relationship between the FDI inflows and GDP, exports and exchange rate.

No.	Author(s)	Description	Methodology	Conclusion
Studies examining the relationship between FDI inflows and/or GDP and/or exports				
Studies relating to developed and developing countries				
1.	De Gregorio (1992), Bende-Nabende and Ford (1998), Olofsdotter (1998), De Mello (1999), Ericsson and Irandoust (2001), Bengoa and Sanchez-Robles (2003), Choe (2003), Li and Liu (2005) and	Effects of the FDI inflows on growth or examine causation from the FDI inflows to growth	Regression estimations and/or co-integration analysis and/or Granger causality test.	All these studies show positive and significant correlation between the FDI inflows and growth.
2.	Johnson (2006)	Investigates the potential of the FDI inflows on host country economic growth	Regression analysis	The FDI inflows enhance economic growth in developing economies but not in developed economies
3.	Carkovic and Levine (2002)	Effects of the FDI inflows on growth	Regression analysis	The FDI inflows do not have any significant influence on economic growth.
Studies relating to developing countries				
4.	Blomstrom et al. (1994), Balasubramanyam et al. (1996), Borensztein et al. (1998), Zhang (2001), Nair-Reichert and Weinhold (2001), Kumar and Pradhan (2002), Basu et al. (2003), Alfaro et al. (2004), Makki and Somwaru (2004), Chowdhury and Mavrotas (2005), Hansen and Rand (2006) and Jajri (2009)	Examine the effects of the FDI inflows on growth/trade/exports or examine the nature of causality between them	Regression estimations and/or co-integration analysis and/or Granger causality test	All these studies conclude that the FDI inflows have positive effect on GDP. However, countries with well-developed financial markets and/or export promoting host countries gain significantly from the FDI and the magnitude of benefits depends upon host country's structural environment.
Studies relating to India				
5.	Sharma (2000)	Factors explaining export growth in India	Simultaneous equation model	The FDI inflows have no significant effect on export performance although its coefficient had a positive sign.
6.	Chakraborty and Basu (2002) and Agrawal (2005)	Linkage between the FDI inflows and growth	VECM and fixed effects model	The FDI inflows play no significant role in the short-run adjustment process of the GDP.
7.	Chakraborty and Nunnenkamp (2006) and Sethi (2007)	Growth implications of the FDI inflows in India	Cointegration test and Granger causality tests	The FDI stocks and output are found to be cointegrated both in short- and long-run. FDI stocks and output are mutually reinforcing for the manufacturing sector. But there was absence of any type of causal relation for the primary sector. Overall, the FDI inflows positively affect the economic growth in India.
8.	Pradhan (2010)	Examine the role of trade openness on the FDI inflows	Regression estimation	The trade openness in India has a significant positive impact on the FDI inflows.
9.	Guru Gharana (2012)	Investigates the causal relationship of GDP with trade and the FDI inflows	ARDL bound testing app. (Auto-regressive distributed lag model)	Export shows short-run as well as long-run Granger causality towards the GDP. But the FDI inflows do not show a short-run Granger causality towards the GDP.
Studies examining the relationship between FDI inflows and exchange rate				
10.	Froot and Stein (1991), Benassy-Quere et al. (2001), Zhang (2001) and Mwega and Ngugi (2006)	Analyse the role of the exchange rate level in the amount of the FDI inflows for the host country	Investigated the inward FDI for the US at sectoral level using regression	The study concluded that the depreciation of the dollar in host country lead to augmentation in the volume of the inward FDI due to decrease investment cost and increase in the wealth of investors.

(*continued*)

Table 1. Continued.

No.	Author(s)	Description	Methodology	Conclusion
11.	Campa (1993)	Analyse the role of the exchange rate level in the amount of the FDI inflows in the US during the 1980s	Using regression analysis	The findings revealed that an appreciation in the US exchange rate stimulated the FDI inflows. The argument was that MNCs search for profits in the host country or local market. If expectant regarding the future profitability, they will escalate the investment in that market in comparison to other available options. Therefore, this model indicated that an appreciation of the host country's currency would increases the inflow of FDI.

Source: Compiled by the authors.

3.1. Unit root testing

Before indulging in econometric modelling using time-series data, we test for stationarity of all the series (Granger, 1981; Granger & Newbold, 1974). The ARDL bounds test is valid for the model where all the variables are integrated of an order zero or one, i.e. I(0) or I(1). For testing stationarity of different series, the study employs the conventional Augmented Dickey–Fuller (ADF) test and the Phillips–Perron (PP) test. At the same time, the study also recognises the effect of presence of structural break in the time series data. Structural break in any time series can occur for number of reasons, e.g. changes in institutional arrangements, policy changes, economic crises and regime shifts. If structural changes are presented in the data generating process, but if not allowed, results may be biased towards the erroneous non-rejection of the non-stationarity hypothesis (Leybourne et al., 2003; Perron, 1989, 1997). India experienced a major transformation in its structural and economical formation during 1991, a year created a history for remarkable changes and evolution. For this purpose, the study has performed Zivot and Andrews (1992) and Perron (1997) unit root testing technique (one-time structural break) for the four variables, i.e. GDP, FDI inflows, export and exchange rate.

3.2. ARDL bound testing approach

Step 1: Examination of long-run cointegration

After determining the suitability of the ARDL approach though unit root testing, the next step is to design a system of unrestricted error correction model (UECM). The UECM is a system of equations using each variable in turn as the dependent variable. It involves just a single-equation set-up, making it simple to implement and interpret. The UECM for this study can be represented as follows:

$$\Delta[\ln\left(X\left(1\right)\right)t] = \beta_o + \sum_{i=1}^{p}\alpha_{1,i}\Delta\ln X\left(1\right)_{t-i} + \sum_{i=0}^{q}\alpha_{2,i}\Delta\ln X\left(2\right)_{t-i} + \sum_{i=0}^{q}\alpha_{3,i}\Delta\ln X\left(3\right)_{t-i} + \sum_{i=0}^{q}\alpha_{4,i}\Delta\ln X\left(4\right)_{t-i} + \lambda_1 X\left(1\right)_{t-i} + \lambda_2 X\left(2\right)_{t-i} + \lambda_3 X\left(3\right)_{t-i} + \lambda_4 X\left(4\right)_{t-i} + \varepsilon_{\alpha t}$$

Here, the λ's are the long run multipliers, β's are the drift terms, α's are the short term dynamic coefficients, Δ is the difference operator and ε's are white noise error terms, that is, independent and identically distributed (i. i. d.) with zero mean, homoscedasticity and no autocorrelation.

Cointegration test: WALD (F) test (Wald, 1943). To test the presence of long-run relationship among the variables of interest (GDP, FDI inflows, export and exchange rate), bound testing procedure under Pesaran et al. (2001) is followed. The bound testing procedure is based on the F-test. The F-test is actually a test of the null hypothesis of no cointegration among the variables against the existence or presence of cointegration among the variables, denoted as:

Null hypothesis: $H_o : \lambda_1 = \lambda_2 = \lambda_2 = \lambda_4 = 0$ (There is no cointegration among the variables),

Alternative hypothesis: $H_a : \lambda_1 \neq \lambda_2 \neq \lambda_3 \neq \lambda_4 \neq 0$ (There is cointegration among the variables).

The ARDL bound testing approach is based on the Wald-test (F-statistic). Two sets of critical values or asymptotic critical value bounds are reported in Pesaran et al. (2001) and Narayan (2005) for various sample sizes assuming that the independent variables are I(d) (where $0 \leq d \leq 1$, that is integrated of order 1 or less). If the calculated F statistics fall outside the two bounds, a conclusive inference can be drawn. If the calculated value exceeds the upper bound of the critical values, then the null hypothesis of no cointegration is rejected. If the calculated values fall below the lower bound of the critical values, then the null hypothesis of no cointegration cannot be rejected. If, however, the calculated F falls within the two bounds, then the test is inconclusive, and further assessment of the accurate order of cointegration has to be performed.

Step 2: Estimation of ARDL models and their robustness

Determination of ARDL model. Once there is establishment or presence of co-integration or long-run relationship among the variables, the study progress towards the estimation of appropriate ARDL (p, q, r, s) model. The selected ARDL (p, q, r, s) model estimates the long-run coefficients and the associated ARDL error correction model for short-run coefficients.

Diagnostic and stability tests. Before proceeding for the determination of long-and short-run coefficient of specified ARDL (p, q, r, s) model, the most important issue addressed in the literature is the stability of the long-run coefficients used to form the error-correction term (ECT) in conjunction with the short-run dynamics. To examine the reliability, the study examines all ARDL (p, q, r, s) models for serial correlation test and heteroscedasticity test. These tests will suggest whether the selected model is suitable or not for further investigation.

For checking the stability of parameters, the cumulative sum (CUSUM) and the cumulative sum of squares (CUSUMSQ) introduced by Brown et al. (1975) are used. The test CUSUM shows if the CUSUM goes out the two critical lines, then the model is not free from instability. Similarly, CUSUMQ, which is based on squared recursive residuals, goes outside the critical lines concludes that the considered model is unstable by nature.

Step 3: Estimation of the long-run ARDL (p, q, r, s) model

Once the model is found to be stable and reliable, the next step is to estimate the conditional ARDL (p, q, r, s) long-run model. Treating one by one each variable as the dependent variable, the augmented ARDL (p, q, r, s) model can be expressed as:

$$\ln\,(Y)_t = C_o + \sum_{i=1}^{p} \delta_1 \ln Y_{t-i} + \sum_{i=0}^{q} \delta_2 \ln X\,(1)_{t-i} + \sum_{i=0}^{r} \delta_3 \ln X\,(2)_{t-i} + \sum_{i=0}^{s} \delta_4 \ln X\,(4)_{t-i} + \mu_t$$

where Y is a dependent variable and independent variables are denoted as $X(1)$, $X(2)$ and $X(3)$, C_o is the constant term and μ_t is a white noise error term. This step involves the selection of optimum lag orders for considered ARDL model (p, q, r, s) within four variables structure. Pesaran and Shin (1999) have shown that the valid asymptotic inferences on short- and long-run parameters made under least squares estimates of an ARDL model (Narayan & Smyth, 2004). As the data is annual, we allow for maximum lag order as 2.

Step 4: The error correction model and short run dynamics

In the next step, we estimate the error correction model. The error correction model results indicate the speed of correction or adjustment of the considered model back to the long-run (said) equilibrium following a short-run shock. So, in the final step, we achieve the short-run dynamic parameters by estimating an error correction model associated with the long-run estimates, i.e. ARDL (p, q, r, s). This is specified as follows:

$$\Delta \ln (Y)_t = C_o + \sum_{i=1}^{p} \beta_1 \Delta \ln Y_{t-i} + \sum_{i=0}^{q} \beta_2 \Delta \ln X(1)_{t-i} + \sum_{i=0}^{r} \beta_3 \Delta \ln X(2)_{t-i} + \sum_{i=0}^{s} \beta_4 \Delta \ln X(4)_{t-i} + \lambda \mathrm{ECM}_{t-1} + \vartheta_t$$

where β_1, β_2, β_3 and β_4 are the short-run dynamic coefficients of the model's convergence back to the given equilibrium. ECM_{t-1} is the error-correction term resulting from the established long-run equilibrium relationship,

and λ is a parameter indicating the speed of adjustment to the equilibrium level after a shock. The sign of the ECM_{t-1} shows negative means a proportion of the disequilibrium is corrected in the next period. The error correction process is thus, a means to reconcile short- and long-run behaviour.

3.3. Granger causality test

The presence of a long-run equilibrium relation among the variables indicates the probability of existence of causality in at least one direction. For this purpose, we apply the Granger causality test (Granger, 1969, 1988) augmented by the ECT. The advantage of using an error correction specification for causality test is that, on the one side, it allows testing for short-run causality through the lagged differenced explanatory variables, and on the other side, the lagged ECM_{t-1}term represents a long-run causality.

In our case, the multivariate p[th] order vector error correction model (VECM) is given by the following equation:

$$\Delta X_t = \varphi + \sum_{i=1}^{n} \theta_i X_{t-i} + \lambda \mathrm{ECM}_{t-1} + \varepsilon_t$$

where X_t is the 4×1 vector of the variables included in the model (GDP, FDI inflows, export and exchange rate), φ is the 4×1 vector of constant terms, θ_i is the 4×4 matrices, which include the interaction coefficients of the variables involved. λ is the 4×1 vector of coefficients for each of the error correction terms, and *et* is the vector of disturbance terms.

4. Results and analysis

The results of the conventional unit root tests, i.e. the ADF, Phillips–Perron (PP) test and unit root tests with structural break, i.e. Zivot and Andrews (1992) and Perron (1997) show that all variables (i.e. GDP, FDI inflows, exports and exchange rate) are non-stationary at level of their logarithmic value shows stationarity properties at their first difference (Refer Tables A1 and A2 in the Appendix). Thus, the maximum order of integration of the variables under study is I(1) and now we can safely employ the ARDL bound testing approach.

4.1. Examination of long-run cointegration

A system of UECM is estimated as follows:

$$\begin{aligned}\Delta[\mathrm{L(GDP)}_t] = {} & \beta_o + \sum_{i=1}^{p} \alpha_{gg,i} \Delta \mathrm{LGDP}_{t-i} + \sum_{i=0}^{q} \alpha_{gf,i} \Delta \mathrm{LFDI}_{t-i} + \sum_{i=0}^{q} \alpha_{ge,i} \Delta \mathrm{LEXPORT}_{t-i} \\ & + \sum_{i=0}^{q} \alpha_{gx,i} \Delta \mathrm{LEXRATE}_{t-i} + \lambda_{gg} \mathrm{GDP}_{t-i} + \lambda_{gf} \mathrm{FDI}_{t-i} + \lambda_{ge} \mathrm{EXPORT}_{t-i} + \lambda_{gx} \mathrm{EXRATE}_{t-i} + \varepsilon_{\alpha t}\end{aligned} \tag{1}$$

$$\begin{aligned}\Delta[\mathrm{L(FDI)}_t] = {} & \beta_o + \sum_{i=1}^{p} \alpha_{ff,i} \Delta \mathrm{LFDI}_{t-i} + \sum_{i=0}^{q} \alpha_{fg,i} \Delta \mathrm{LGDP}_{t-i} + \sum_{i=0}^{q} \alpha_{fe,i} \Delta \mathrm{LEXPORT}_{t-i} \\ & + \sum_{i=0}^{q} \alpha_{fx,i} \Delta \mathrm{LEXRATE}_{t-i} + \lambda_{ff} \mathrm{FDI}_{t-i} + \lambda_{fg} \mathrm{GDP}_{t-i} + \lambda_{fe} \mathrm{EXPORT}_{t-i} + \lambda_{fx} \mathrm{EXRATE}_{t-i} + \varepsilon_{\alpha t}\end{aligned} \tag{2}$$

$$\begin{aligned}\Delta[\mathrm{L(EXPORT)}_t] = {} & \beta_o + \sum_{i=1}^{p} \alpha_{ee,i} \Delta \mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{q} \alpha_{eg,i} \Delta \mathrm{LGDP}_{t-i} + \sum_{i=0}^{q} \alpha_{ef,i} \Delta \mathrm{LFDI}_{t-i} \\ & + \sum_{i=0}^{q} \alpha_{ex,i} \Delta \mathrm{LEXRATE}_{t-i} + \lambda_{ef} \mathrm{FDI}_{t-i} + \lambda_{eg} \mathrm{GDP}_{t-i} + \lambda_{ee} \mathrm{EXPORT}_{t-i} + \lambda_{ex} \mathrm{EXRATE}_{t-i} + \varepsilon_{\alpha t}\end{aligned} \tag{3}$$

$$\begin{aligned}\Delta[\mathrm{L(EXRATE)}_t] = {} & \beta_o + \sum_{i=1}^{p} \alpha_{xx,i} \Delta \mathrm{LEXRATE}_{t-i} + \sum_{i=0}^{q} \alpha_{xg,i} \Delta \mathrm{LGDP}_{t-i} + \sum_{i=0}^{q} \alpha_{xf,i} \Delta \mathrm{LFDI}_{t-i} \\ & + \sum_{i=0}^{q} \alpha_{xe,i} \Delta \mathrm{LEXPORT}_{t-i} + \lambda_{xf} \mathrm{FDI}_{t-i} + \lambda_{xg} \mathrm{GDP}_{t-i} + \lambda_{xe} \mathrm{EXPORT}_{t-i} + \lambda_{xx} \mathrm{EXRATE}_{t-i} + \varepsilon_{\alpha t}\end{aligned} \tag{4}$$

The optimal lag order on the first-difference variables in Equations (1–4) can be obtained from unrestricted vector autoregression (UVAR) by means of Akaike and/or Schwarz criteria. Bahmani-Oskooee and Bohl (2000) and Bahmani-Oskooee and Ng (2002) have, however, shown that the results of this stage are sensitive to the order of

VAR. We allow for maximum lag order of 2, taking in consideration that our model consists of annual data. After the examination through UVAR model, the optimal number of lags for GDP as a dependent variable is 1, for the FDI inflows is 0, for exports is 1 and for exchange rate is also 1 (Refer Table A3, Appendix).

Bound test for cointegration: WALD (F) statistics

The next step in the ARDL bounds testing approach is to estimate equations by ordinary least squares method (OLS), for testing the existence of a long-run relationship among the variables, by conducting an F-test for the joint significance of the coefficients of the lagged levels of the variables involved. The null hypotheses for no cointegration are:

$$H_{o1}: \lambda_{gg} = \lambda_{gf} = \lambda_{ge} = \lambda_{gx} = 0$$
$$H_{a1}: \lambda_{gg} \neq 0, \lambda_{gf} \neq 0, \lambda_{ge} \neq 0, \lambda_{gx} \neq 0 \quad \text{(Hypothesis 1)}$$

$$H_{o2}: \lambda_{fg} = \lambda_{ff} = \lambda_{fe} = \lambda_{fx} = 0$$
$$H_{a2}: \lambda_{fg} \neq 0, \lambda_{ff} \neq 0, \lambda_{fe} \neq 0, \lambda_{fx} \neq 0 \quad \text{(Hypothesis 2)}$$

$$H_{o3}: \lambda_{eg} = \lambda_{ef} = \lambda_{ee} = \lambda_{ex} = 0$$
$$H_{a3}: \lambda_{eg} \neq 0, \lambda_{ef} \neq 0, \lambda_{ee} \neq 0, \lambda_{ex} \neq 0 \quad \text{(Hypothesis 3)}$$

$$H_{o4}: \lambda_{xg} = \lambda_{xf} = \lambda_{xe} = \lambda_{xx} = 0$$
$$H_{a4}: \lambda_{xg} \neq 0, \lambda_{xf} \neq 0, \lambda_{xe} \neq 0, \lambda_{xx} \neq 0 \quad \text{(Hypothesis 4)}$$

Table 2 reports the results of the calculated F statistics when consider each variable as a dependent variable (normalised) under the ARDL-OLS regression considering F-test critical values specified by Pesaran et al. (2001) as well as the Narayan (2005).

With the GDP as a dependent variable, the calculated F-statistics $F(\text{GDP}) = 4.471977$, is higher than the upper bound critical value specified by Pesaran et al. (2001) and Narayan (2005) at 5% significance level. Thus, the null hypothesis of no cointegration is rejected. For the FDI inflows as a dependent variable, the calculated F-statistics $F(\text{FDI}) = 4.131014$, which is also above the upper bound critical value specified by Pesaran et al. (2001) at the 5% significance level, but marginally below the upper critical value 4.194 by Narayan (2005). So, considering both the test critical value bounds, we conclude that there is presence of long run relationship among variable taking the FDI inflows as a dependent variable.

Taking exports as a dependent variable, the calculated F-statistics shows $F(\text{EXPORT}) = 4.256390$, which is higher than the upper bound critical value specified by both Pesaran et al. (2001) and Narayan (2005). Therefore, one can reject the null hypotheses of no cointegration, implying a long-run cointegration for export as a dependent variable. Then looking at exchange rate as a dependent variable, the calculated F-statistics $F(\text{EXRATE}) = 2.993004$, which lies between the upper bound critical value 3.67 I and lower bound critical value 2.79 as specified by Pesaran et al. (2001) and thereby showing inconclusive results. Further, the F-statistics $F(\text{EXRATE})$ value falls below lower level critical value bounds, i.e. 3.164 for Narayan (2005), thus showing no cointegration results. So, we conclude that we cannot reject the null hypotheses of no cointegration, implying that there is no long-term relationship between variables taking exchange rate as a dependent variable.

Hence, from the overall analysis, it can be seen that for the variables GDP, the FDI inflows and exports (except exchange rate), the null hypothesis of no cointegration is rejected, indicating there is a stable long-run cointegration relationship among the variables. The fact that the variables are cointegrated implies that there is some correction process in the short-term, which prevents errors in long-term interaction to become bigger.

Table 2. Bound test for cointegration: WALD (F) statistics.

Dependent variable	F-statistics	Independent variable	F-statistics value	Probability	Outcome
L(GDP)	F(*GDP*)	L(FDI), L(EXPORT), L(EXRATE)	4.471077**	0.0103	Cointegration
L(FDI)	F(*FDI*)	L(GDP), L(EXPORT), L(EXRATE)	4.131014**	0.0115	Cointegration
L(EXPORT)	F(*EXPORT*)	L(GDP), L(FDI), L(EXRATE)	4.256390**	0.0126	Cointegration
L(EXRATE)	F(*EXRATE*)	L(GDP), L(FDI), L(EXPORT)	2.993004	0.0450	No cointegration

Critical values bounds for F-statistics are obtained from both Pesaran et al. (2001) and Narayan (2005).
Asymptotic critical values bounds from Table C I (ii) case II (Restricted intercept and no trend with three regressors as K = 3) in Pesaran et al. (2001). Lower bound I (0) = 2.79 and upper bound I (1) = 3.67 at 5% significance level.
According to Narayan (2005), the existing critical values reported in Pesaran et al. (2001) are mainly used for the large sample size data. Narayan (2005) provides a set of critical values for relatively small sample size ranging from 30 to 80 observations.
Narayan (2005) case II results are slightly different from Pesaran et al. (2001). Values for lower bound = 3.164 and for upper bound = 4.194 with restricted intercept and no trend for 35 observations with three regressors as K = 3 at 5% significance level.
** signifies statistically significant at 5% level.

4.2. Estimation of ARDL models and robustness

Once the existence of co-integration relationship among the considered variables is established, the following equations are estimated for the long-run coefficients of the selected ARDL model and the associated error correction model for short-run coefficients. Before moving for the estimation of long- and short-run parameters, we check each of the three ARDL (p, q, r, s) models for their reliability and stability.

Diagnostic tests: Considering serial correlation test (Appendix Table A4), the probability corresponding to F-statistic for each of three variables (i.e. GDP, FDI inflows, exports and exchange rate) is more than 1, 5 and 10%. Therefore, we cannot reject our null hypothesis at 1, 5 or 10% significance level meaning that we accept our null hypothesis of no serial correlation. In the results for heteroscedasticity test (Appendix Table A4), the probability corresponding to F-statistic for each of three variables (i.e. GDP, FDI inflows, exports and exchange rate) is also more than 1, 5 and 10%. Hence, here we cannot reject the null hypothesis at 1, 5 or 10% significance level meaning that we accept the null hypothesis of no heteroscedasticity.

Stability tests: The CUSUM and CUSUMSQ plots for the estimated ARDL (p, q, r, s) models for GDP, FDI inflows and exports as a dependent variable are shown in Figures 1–3(a,b), respectively in the Appendix. The assessment of plots shows that CUSUM and CUSUMSQ statistics (for GDP, FDI inflows and exports as a dependent variable) lies between the 5% critical bounds lines implying that short- and long-run coefficients of the selected ARDL (p, q, r, s) in the given model are stable. Therefore, one can conclude that the coefficients in the given regression are constant. Hereby the study concludes that the above models are perfect (free from the problem of serial correlation, heteroscedasticity or instability) and can be used for further investigation, and/or forecasting.

4.3. Estimation of the ARDL model

Treating each of three variables as the dependent variable, the augmented ARDL (p, q, r, s) long-run model can be shown as:

$$L(\mathrm{GDP})_t = \mathrm{C}_o + \sum_{i=1}^{p}\delta_1 \mathrm{LGDP}_{t-i} + \sum_{i=0}^{q}\delta_2 \mathrm{LFDI}_{t-i} + \sum_{i=0}^{r}\delta_3 \mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{s}\delta_4 \mathrm{LEXRATE}_{t-i} + \mu_t \tag{5}$$

$$L(\mathrm{FDI})t = \mathrm{C}_o + \sum_{i=1}^{p}\delta_1 \mathrm{LFDI}_{t-i} + \sum_{i=0}^{q}\delta_2 \mathrm{LGDP}_{t-i} + \sum_{i=0}^{r}\delta_3 \mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{s}\delta_4 \mathrm{LEXRATE}_{t-i} + \mu_t \tag{6}$$

$$L(\mathrm{EXPORT})_t = \mathrm{C}_o + \sum_{i=1}^{p}\delta_1 \mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{q}\delta_2 \mathrm{LGDP}_{t-i} + \sum_{i=0}^{r}\delta_3 \mathrm{LFDI}_{t-i} + \sum_{i=0}^{s}\delta_4 \mathrm{LEXRATE}_{t-i} + \mu_t \tag{7}$$

So, after the estimation of the ARDL (p, q, r, s) long-run multipliers, the next step is to estimate the short-run dynamic coefficients via the following error correction model:

$$\Delta L(\mathrm{GDP})_t = \mathrm{C}_o + \sum_{i=1}^{p}\beta_2 \Delta\mathrm{LGDP}_{t-i} + \sum_{i=0}^{q}\beta_2 \Delta\mathrm{LFDI}_{t-i} + \sum_{i=0}^{r}\beta_3 \Delta\mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{s}\beta_4 \Delta\mathrm{LEXRATE}_{t-i} + \lambda \mathrm{ECM}_{t-1} + \vartheta_t \tag{8}$$

$$\Delta L(\mathrm{FDI})_t = \mathrm{C}_o + \sum_{i=1}^{p}\beta_2 \Delta\mathrm{LFDI}_{t-i} + \sum_{i=0}^{q}\beta_2 \Delta\mathrm{LGDP}_{t-i} + \sum_{i=0}^{r}\beta_3 \Delta\mathrm{LEXPORT}_{t-i} + \sum_{i=0}^{s}\beta_4 \Delta\mathrm{LEXRATE}_{t-i} + \lambda \mathrm{ECM}_{t-1} + \vartheta_t \tag{9}$$

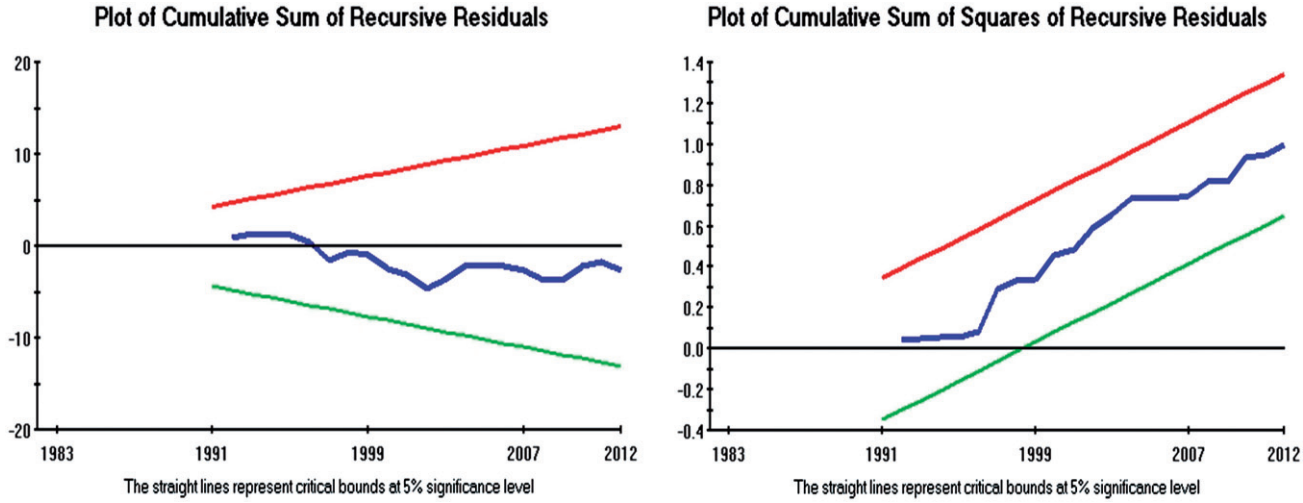


Figure 1. GDP as a dependent variable (a) CUSUM and (b) CUSUMSQ based on SBC [ARDL (1,2,0,1)].

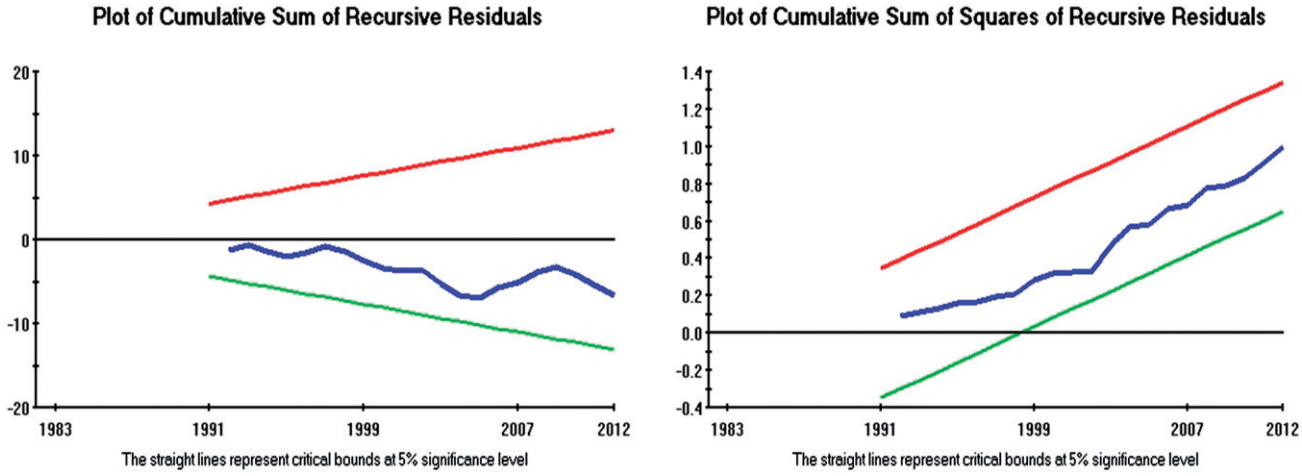


Figure 2. FDI as a dependent variable (a) CUSUM and (b) CUSUMSQ based on SBC [ARDL (2,0,0,2)].

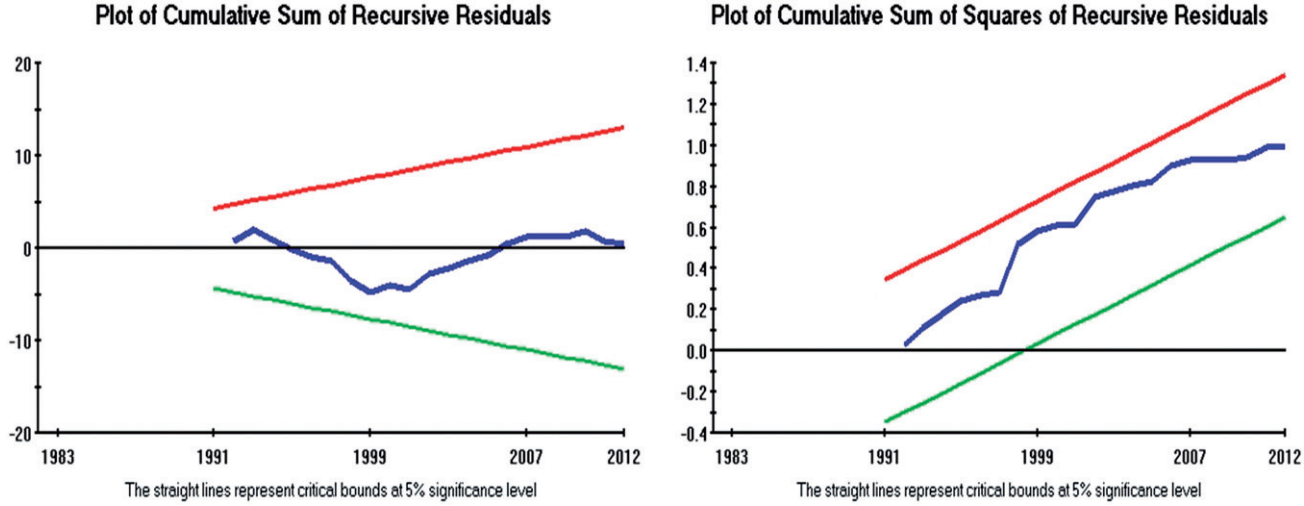


Figure 3. Exports as a dependent variable (a) CUSUM and (b) CUSUMSQ based on SBC [ARDL (1,1,0,2)].

$$\Delta L(\text{EXPORT})_t = C_o + \sum_{i=1}^{p} \beta_2 \Delta \text{LEXPORT}_{t-i} + \sum_{i=0}^{q} \beta_2 \Delta \text{LGDP}_{t-i} + \sum_{i=0}^{r} \beta_3 \Delta \text{LFDI}_{t-i} + \sum_{i=0}^{s} \beta_4 \Delta \text{LEXRATE}_{t-i} + \lambda \text{ECM}_{t-1} + \vartheta_t \quad (10)$$

GDP as dependent variable

Long-run estimates of ADRL process. Equation (5) is estimated using the following ARDL (p, q, r, s) specification. The results obtained by normalising on the GDP, in the long-run are reported in Table 3. In Table 3, the estimated coefficients of the long-run relationship show that export, exchange rate and the FDI inflow has a significant impact on the GDP. A 1% increase in exports leads to approximately 0. 71149% increase in the GDP, other things being equal at 1% level of significance. According to the proponents of export-led growth (ELG) hypothesis, exports can perform the function of an 'engine of growth'. The association between exports and economic growth is often attributed to positive externalities for the domestic economy arising from participation in world markets, re-allocation of existing resources, economies of scale and various labour-specialisation effects (Bahmani-Oskooee & Alse, 1993; Bhagwati, 1978; Henriques & Sadorsky, 1996; Krueger, 1978; Serletis, 1992).

The estimated coefficient of exchange rate is negative (i.e. = −0.55989) and statistically significant at 1% level, implying that the appreciation of domestic currency is more effective in stimulating the growth prospects in India over a period of time, assuming other things being equal. The reason behind this can be attributable to cheaper imports because appreciating rupee is helpful for those industries and sectors like auto, engineering and aviation companies, which rely heavily on imported inputs. Further, the FDI inflows also positively affect the GDP, significant at 5% level keeping other thing being equal. A considerable positive effect shows that the FDI inflow to developing nations like India provides a supplement for domestic credit and introduces new technology, along with providing more employment opportunities for the host country. Chakraborty and Nunnenkamp (2008) and Sethi (2013) indicated that booming FDI in post-reform India is widely believed to promote economic growth as a whole.

Short-run dynamics of ADRL process. The results of short-run dynamics and their coefficients associated with the long-run relationships obtained from the ARDL-ECM equation are presented in Table 4.

Results of the ECM model show that the coefficient of LEXPORT is positively associated with the GDP and significant at 1%. Similarly, the coefficient of LFDI shows positive relation with the GDP enhancement and it is significant at 10% level shows positive momentous in short-run. Now, in a short-run mechanism, exchange rate reveals a dynamic stance. The coefficient of current year's exchange rate is negatively related to current year GDP. But last year's exchange rate is positively and significantly related to current year's GDP, other things being constant. This indicates towards an idiosyncratic cyclic nature of adjustment between exchange rate and GDP. The positive magnitude of exchange rate of last year means that appreciating foreign currency in terms of rupee provides advantageous platform for exporters to gain out of that. IN the meantime, the falling rupee value makes Indian domestic goods and services cheaper for foreign buyers. So, it increases the demand of Indian goods and services in other countries. The foreign tourists can also enjoy cost effective scenario in this case. This would all affect the GDP in positive sense by increased production, having new employment opportunities. This is shown in the coefficient of dLEXRATE1; a 1% depreciation of domestic currency increased the GDP by 0.13041% in short-run. Now,

Table 3. Estimated long-run coefficients: GDP as endogenous variable.

Dependent variable is LGDP			
Estimated long-run coefficients using ARDL approach			
ARDL(1,2,0,1) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
LEXRATE	−0.55989	0.083671	−6.6916 [.000]*
LEXPORT	0.71149	0.04439	16.0280 [.000]*
LFDI	0.068937	0.028795	2.3941 [.026]**
C	7.3462	0.36169	20.3108 [.000]*

* and ** signifies statistically significant at 1% and 5% level respectively.

Table 4. Error correction representation: GDP as endogenous variable.

Dependent Variable is dLGDP			
Error correction representation for selected ARDL model			
ARDL(1,2,0,1) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
dLEXRATE	−0.86055	0.065415	−13.1551 [.000]*
dLEXRATE1	0.13041	0.05825	2.2388 [.035]**
dLEXPORT	0.17647	0.035203	5.0131 [.000]*
dLFDI	0.0031605	0.0060017	0.52661 [.603]***
ECM (−1)	−0.24804	0.053256	−4.6575 [.000]*
ECM = LGDP + 0.55989 × LEXRATE − 0.71149 × LEXPORT − 0.068937 × LFDI − 7.3462 × C			
R-squared = 0.95969		R-Bar-squared = 0.94687	
S.E. of regression = 0.018574		F-stat = 104.7675 [.000]	
DW-statistic = 1.8996			

*, ** and *** signifies statistically significant at 1%, 5% and 10% level respectively.

increased exports and larger capital inflows slowly and suddenly start affecting the exchange rate scenario, keeping other things constant. Moving from last year to current year's state of affairs, things start rectifying themselves and pushing towards a more pragmatic wave. In this way, depreciating domestic currency poses a menace to country in general. Reserve Bank of India (RBI) tries to control that fall by making use of its direct and indirect instruments of monetary policy and government interfere in the market by way of bonds and other desirable routes. As a result, domestic currency starts reviving itself as of this push mechanism. Now, appreciating rupee raises the confidence of importers to fulfil their requirements. By all this, ultimately the GDP goes upwards and the same is shown in the coefficient of dLEXRATE.

The coefficient of ECT is significant at 1% level and its value is estimated as −0.24, which implies that the speed of adjustment to equilibrium after a shock is relatively fair. Approximately 24% of disequilibria from the previous year's shock converge back to the long-run equilibrium in the current year. The error correction model for GDP as a dependent variable is given by:

$$\text{ECM} = \text{LGDP} + 0.55989 * \text{LEXRATE} - 0.71149 * \text{LEXPORT} - 0.068937 * \text{LFDI} - 7.3462 * \text{C}$$

Furthermore, the value of adjusted $R^2 = 0.94$ explains 94% goodness of fit of the above said model or in other words 94% variation in dependent variable, i.e. the GDP is explained by independent variables in the given model. The value of F-stat is highly significant implying that the overall model is robust.

FDI inflows as dependent variable

Long-run estimates of ADRL process. The results of long-run parameters are presented in Table 5. Firstly, the empirical results reveal that a 1% increase in exchange rate leads to 3.2994% increase in the FDI inflows, assuming other things being equal. This means that depreciation of domestic currency has a positive impact on inflow of funds in India. Depreciation of the host country's currency against the home currency increases the relative wealth of foreigners and this may increase the attractiveness of the host country. The studies by Froot and Stein (1991), Kogut and Chung (1996), GoldBerg and Klien (1998), Kiyota and Urata (2004) and Yao (2006) argue that depreciating host currency leads to increase the FDI inflow. The magnitude of exports shows that in long-run there is negative but weak relationship exports and the FDI inflows. This shows that the relation is statistically insignificant that means that exports from the host country are not considered as decisive criteria by foreign investors to invest in a country like India.

The positive and significant coefficient of the GDP shows that 1% increase in GDP leads to 3.7653% increase in the FDI inflows. The result shows that Indian GDP is one of the most important causes of increased level of the FDI inflows. The GDP shows a progress graph of country over a period of time and foreign investors while deciding upon their investment destination's priority, must consider the GDP as one of the most important vital benchmark. Further, Miankhel et al. (2009) found that it is the GDP growth in the long-run that attracts the FDI in India. The results of the short-run dynamic coefficients associated with the long-run relationships are tabulated in Table 6.

The coefficient of lag of the FDI was positive and significant, reflecting the influence of agglomeration effect on foreign investors' decision making with respect to the FDI. This apparent trend shows that inflows from year

Table 5. Estimated long-run coefficients: FDI inflows as endogenous variable.

Dependent Variable is LFDI			
Estimated long-run coefficients using ARDL approach			
ARDL (2,0,0,2) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
LEXRATE	3.2994	0.55448	5.9504 [.000]*
LEXPORT	−1.7635	1.0649	−1.6560 [.112]
LGDP	3.7653	1.4208	2.6500 [.015]**
C	−35.098	9.155	−3.8338 [.001]*

* and ** signifies statistically significant at 1% and 5% level respectively.

Table 6. Error correction representation: FDI inflows as endogenous variable.

Dependent Variable is dLFDI			
Error correction representation for selected ARDL model			
ARDL(2,0,0,2) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
dLFDI1	0.3965	0.1602	2.4751 [.021]**
dLEXRATE	4.2283	0.99725	4.2400 [.000]*
dLEXPORT	−2.2601	1.4473	−1.5615 [.132]
dLGDP	7.6911	2.8999	2.6522 [.014]*
dLGDP1	4.013	1.7749	2.2610 [.034]**
ECM (−1)	−1.2816	0.21849	−5.8654 [.000]*
ECM = LFDI − 3.2994 × LEXRATE +1.7635 × LEXPORT − 3.7653 × LGDP + 35.0980 × C			
R-squared = 0.62933		R-Bar-squared = 0.51139	
S.E. of regression = 0.56864		F-stat = 6.2254 [.001]	
DW-statistic = 1.8932			

* and ** signifies statistically significant at 1% and 5% level respectively.

to year have a progressive and positive impact. Again, the coefficient of exchange rate dLEXRATE indicates that a 1% increase in exchange rate leads to 4.2283% increase in the FDI inflows. This means that short-run drop in domestic currency rate has an affirmative effect on the FDI inflows in India.

Now, looking at coefficient of export (i.e. −2.2601) is negatively related with the FDI inflows in the short-run but again this relationship is not significant. The coefficient of both dLGDP and dLGDP1 has a significant and positive impact on the FDI Inflows. This means that GDP of current year and last year both influence positively and significantly on the FDI inflows.

The estimated coefficient of the ECM (−1) is equal to −1.2816 showing a relatively quick speed of adjustment back to the long-run equilibrium after a shock. The error correction model for FDI inflows as a dependent variable is given by:

$$\text{ECM} = \text{LFDI} - 3.2994 * \text{LEXRATE} + 1.7635 * \text{LEXPORT} - 3.7653 * \text{LGDP} + 35.0980 * \text{C}$$

Correspondingly, the value of adjusted $R^2 = 0.51$ explains 51% goodness of fit of the model means that 51% of the proportion of total variance in the FDI inflows is explained by the considered model.

Exports as dependent variable

Considering exports as a dependent variable (Equation (7)), the normalised long-run coefficients of the selected ARDL(1,1,0,2) are given in Table 7.

Table 7 displays the negative relationship between the FDI inflows and exports. This means that a 1% increase in the FDI inflows leads to 0.12041% decrease in exports assuming other things being equal. Though, the role of the FDI inflows in promoting export is always remains a divisive matter and this fundamentally depends upon the motive for such investment (Dunning, 1998). If the motive for the FDI is to bypass the trade barriers of the host countries, to gain access to large overseas market or seeks to serve the host market, it may not promote exports. According to World Investment Prospects Survey (WIP Survey 2013–2015, United Nations, 2013), India ranked number third among the TNCs' top prospective host economies. This means that foreign investors consider India

Table 7. Estimated long run coefficients: Exports as endogenous variable.

Dependent variable EXPORT			
Estimated long-run coefficients using ARDL approach			
ARDL(1,1,0,2) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
LFDI	−0.12041	0.056661	−2.1251 [.045]**
LEXRATE	0.79311	0.16519	4.8012 [.000]*
LGDP	1.4689	0.10384	14.1461 [.000]*
C	−10.5488	1.3358	−7.8973 [.000]*

* and ** signifies statistically significant at 1% and 5% level respectively.

among the top investment priority destination. Foreign investors primarily invest in India to explore new market. It points towards the likelihood that India attracts more of the market seeking FDI. They set up the subsidiaries and branches in India with intention to cater host country's market, not for exporting purpose.

Moreover the domestic merchandises, products and services earlier used for export purpose are now being consumed by these gigantic brands. Thus, the market-seeking FDI might lead to higher GDP, but less of trade. In the meantime, increased competition in the domestic market at times throws out some of the national players, as a result of non-survival. This results in slowing down the export mechanism in India. However, UNCTAD (1998) is sceptical about the positive contribution of the FDI inflows on manufacturing export performance as it opinions that consumption goods and capital not accessible locally can be imported, and thereby profits remitted, thus cutting into the export earnings generated.

Now, the coefficient of exchange rate shows that 1% increase in exchange rate leads to 0.79311% increase in exports, other things being equal implies that the depreciation of currency is more effective in stimulating the growth of exports in India. In other words, a falling rupee results in greater profitable margin for exporters. Export-oriented sectors that benefits out of rupee depreciation include software services, pharmaceutical and aquaculture.

The table results show that the GDP exerts positive and significant effect on real exports in the long-run, assuming other things being equal. This suggests that higher production of goods and services and commodities in different sectors of the economy encourages greater exports from India. The notion behind the theory of this economic growth promotes export growth, which argues that gain in productivity overtime, gives rise to comparative advantage in certain sectors of economy. This eventually leads to export growth as a whole (Dhawan & Biswal, 1999; Ghatak & Price, 1997).

Moreover, the results pertaining with short-run coefficients associated with the long-run relationships obtained from the ECM equation are given in Table 8.

The coefficient of dLFDI is negatively related with export as a dependent variable but this relationship is statistically insignificant, which means that in short-run FDI inflows have no significant impact on exports. The coefficient, i.e. dLEXRATE implies that depreciation of domestic currency has a positive and significant effect on exports in the short-run. This means 1% depreciation in rupee increased exports by 0.42215% keeping other things being equal. A cheaper rupee opportunity, if grabbed appropriately, can provide Indian companies with additional revenue streams in an otherwise tougher demand environment.

The coefficient of both dLGDP and dLGDP1 has a significant and positive impact on exports, assuming other things being equal. This means that GDP of current year and last year both influence positively and significantly the exports growth in India.

The estimated error correction term/coefficient (ECT), i.e. −0.53227 is negative and significant at 1% level ensuring that the adjustment process from the short-run deviation is quite temperate. More precisely, this indicates that 53% of the disequilibrium in exports from the previous period's shock is converged back to the long-run equilibrium in this period.

The error correction model considering exports as a dependent variable is specified as follows:

$$\mathrm{ECM} = \mathrm{LEXPORT} + 0.12041 * \mathrm{LFDI} - 0.79311 * \mathrm{LEXRATE} - 1.4689 * \mathrm{LGDP} + 10.5488 * \mathrm{C}$$

The value of adjusted $R^2 = 0.61$ signifies 61% goodness of fit, means that 61% of the proportion of total variance in the exports in India is explained by the considered model. F-stat shows that the above model is best fitted.

Table 8. Error correction representation: Exports as endogenous variable.

Dependent variable dLEXPORT			
Error correction representation for selected ARDL model			
ARDL(1,1,0,2) selected based on Schwarz Bayesian Criterion			
Regressor	Coefficient	Standard error	T-ratio [Prob.]
dLFDI	−0.025124	0.021451	−1.1712 [.253]
dLEXRATE	0.42215	0.10757	3.9245 [.001]*
dLGDP	1.4484	0.22476	6.4444 [.000]*
dLGDP1	0.43203	0.20119	2.1474 [.042]**
ECM (−1)	−0.53227	0.14717	−3.6167 [.001]*
ECM = LEXPORT + 0.12041 × LFDI − 0.79311 × LEXRATE − 1.4689 × LGDP + 10.5488 × C			
R-squared = 0.70695		R-Bar-squared = 0.61371	
S.E. of regression = 0.066657		F-stat = 10.6146 [.000]	
DW-statistic = 1.7007			

* and ** signifies statistically significant at 1% and 5% level respectively.

4.4. Granger causality test

The existence of cointegration relationship indicates that there are long-run relationships among the examined variables, and this also indicates that there must be Granger causality among them in at least one direction (Engle & Granger, 1987). Thus, in order to determine the short- and long-run causality, we performed the augmented form of Granger causality test using the error-correction-based Granger causality model.

It is important to remember that only for those equations where long-run relationship is already established (i.e. GDP, FDI inflows and exports), an ECT is estimated (Halicioglu, 2009; Narayan and Smyth, 2008; Odhiambo, 2009; Uddin et al., 2011). However, Granger (1988) noted that, if a set of variables are cointegrated, the short- and long-run causality cannot be captured by the standard first difference VAR model. Therefore, Granger causality test require the inclusion of a lagged error-correction term within a VECM in order to capture the short-run deviations of the series from their long-run equilibrium relationship (Narayan & Smyth, 2004). Accordingly, Granger causality analysis within the VECM involves estimating the following models:

$$\Delta L\,(\mathrm{GDP})_t = \alpha_1 + \sum_{i=1}^{p}\beta_i\Delta\mathrm{LGDP}_{t-i} + \sum_{i=1}^{p}\vartheta_i\Delta\mathrm{LFDI}_{t-i} + \sum_{i=1}^{p}\kappa_i\Delta\mathrm{LEXPORT}_{t-i} + \sum_{i=1}^{p}\theta_i\Delta\mathrm{LEXRATE}_{t-i} + \psi_1\mathrm{ECT}_{t-1} + \upsilon_{1t} \tag{11}$$

$$\Delta L\,(\mathrm{FDI})_t = \alpha_2 + \sum_{i=1}^{p}\vartheta_i\Delta\mathrm{LFDI}_{t-i} + \sum_{i=1}^{p}\beta_i\Delta\mathrm{LGDP}_{t-i} + \sum_{i=1}^{p}\kappa_i\Delta\mathrm{LEXPORT}_{t-i} + \sum_{i=1}^{p}\theta_i\Delta\mathrm{LEXRATE}_{t-i} + \psi_2\mathrm{ECT}_{t-1} + \upsilon_{2t} \tag{12}$$

$$\Delta L\,(\mathrm{EXPORT})_t = \alpha_3 + \sum_{i=1}^{p}\kappa_i\Delta\mathrm{LEXPORT}_{t-i} + \sum_{i=1}^{p}\beta_i\Delta\mathrm{LGDP}_{t-i} + \sum_{i=1}^{p}\vartheta_i\Delta\mathrm{LFDI}_{t-i} + \sum_{i=1}^{p}\theta_i\Delta\mathrm{LEXRATE}_{t-i} + \psi_3\mathrm{ECT}_{t-1} + \upsilon_{3t} \tag{13}$$

The results of Granger causality are presented in Table 9. Since the variables are cointegrated, the direction of causality can be divided into two parts: short- and long-run causation. The one period lagged error-correction term ECT_{t-1} represents the long-run causality and the lagged differenced explanatory variables represent the short-run causality.

The GDP equation shows that there is a presence of uni-directional causality, running from exchange rate to the GDP significant at 1% significance level in the short-run. We also observe a bidirectional causal relationship between the GDP and export indicating that both Grangers cause each other in the short run. These results are in confirmation of previous studies (Dutt & Ghosh, 1994; Khalafalla & Webb, 2001; Shan & Sun, 1998; Thornton, 1996). Looking at the long-run causality, these empirical results suggest that all the ECT_{t-1} coefficients are negatively signed and statistically significant at 1% significance level. This implies that long-run causality running from

Table 9. Granger causality test results.

	Type of Granger causality test				
	Short-run				Long-run
Dependent variable	d(LGDP)	d(LFDI)	d(LEXPORT)	d(LEXRATE)	ECM_{t-1}
d(LGDP)	–	−0.0010	0.1549	−0.8960	−0.187896
		[0.8469]	[0.0026]*	[0.0000]*	[0.0004]*
d(LFDI)	0.6114	–	−1.8451	−3.3896	−1.175383
	[0.9086]		[0.2137]	[0.5262]	[0.0000]*
d(LEXPORT)	1.4632	−0.0248	–	0.4804	−0.522085
	[0.0189]**	[0.2180]		[0.4630]	[0.0008]*

Figures in parentheses indicate *p* values.
* and ** signifies statistically significant at 1% and 5% level respectively.

the explanatory variables (i.e. FDI inflows, GDP, exports and exchange rate) specified in the model towards each of the dependent variable, i.e. the FDI inflows, GDP and export. The results of causation are in conformance with the bound-testing results.

5. Conclusion and policy implications

Using the most recent and robust ARDL (bound testing) approach to cointegration proposed by Pesaran et al. (2001), this study concludes that there is a strong evidence of long-run relationship between GDP, FDI inflows and exports for the case of India. However, there is lack of evidence of long-run cointegration with exchange rate as a dependent variable. Examining long-run and short-run parameters, the results indicate bilateral positive and significant relationship between the FDI inflows and GDP; and GDP and exports in both the long-run and short-run. In contrast, the results indicate negative impact of the FDI inflows on exports in the long-run. Moreover, exchange rate appreciation has a positive impact on the FDI inflows and exports, but negative effect on the GDP in the long-run. Looking at the ECT, the models signify a fairly quick speed of adjustment to the equilibrium following a shock. Furthermore, according to Granger causality test, the study observes bidirectional causal relationship between the GDP and exports and uni-directional causality from exchange rate to the GDP in the short-run. Additionally, the results confirm long-run causality running from the explanatory variables specified in the model towards the respective dependent variables. With the above results in mind, it can be said that sound macroeconomic fundamentals coupled with proficient capital market raises business confidence among investors and contributes positively to the FDI inflows.

From the results, it can be seen that higher amount of FDI inflows leads to increase in the GDP but it adversely affects export growth in the long-run. The main reason attributable to this may be that India largely attracts the market seeking FDI. Therefore, there is a need for the government to pay attention on devising policies aimed at seeking export-oriented FDI. The FDI reforms should include policies that help remove impediments for export-oriented sectors and should attract MNE's to locate more of efficiency seeking FDI in India. Paudel (2014) argues that India's major export performance originates from the manufacturing sector itself. Therefore, the FDI in this sector should be made more open and liberal. India should capitalise on its advantages such as vast pool of technical professionals and skilled labour, and advanced capital markets to attract a greater proportion of efficiency seeking FDI (Kumar, 1998). Government should focus upon the synchronisation of the FDI policies with exports, so as to increase both the GDP growth rate as well as export performance.

The study found that appreciation of rupee leads to increase in the GDP but decrease in the FDI inflows in long-run, other things being equal. In the other words, depreciation of rupee raises the level of the FDI inflows for India. This may not be a healthy trend as weakening of domestic currency adversely affects a nation's international standing. Further, depreciation in currency over a long period of time could hamper investor confidence regarding future profitability and may eventually become a deterrent to the FDI. Thus, the government policies should aim at developing a less volatile exchange rate system that could facilitate the value certainty of future profits for prospective investors.

Another long-term relationship emerged from the results is the one between exports and economic growth via the GDP growth. Export promotion has a positive influence on growth in the GDP, thereby confirming the advantages of an ELG strategy for India. However, the growth in export over the years has been sluggish. Thus, there is a need for an effective export policy; a standardised export agency should be put in place to monitor the

exportable units, their quality and standardisation of products and services. The policymakers should contemplate increasing the FDI limits in those sectors that account for a large part of the export basket. To sum up, India needs to establish a transparent, liberal and effective FDI policy framework along with strengthening human and institutional capabilities to implement them, so as to achieve the desired effects of the FDI inflows.

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Appendix

Table A1. Unit root testing without structural break.

	ADF Test		PP Test		
Variables	t-stat.: I(0)	t-stat.: I(1)	t-stat.: I(0)	t-stat.: I(1)	Result
GDP	1.695619 (.999)	−4.229645* (.002)	1.695619 (.999)	−4.222067* (.002)	Series: I(1)
FDI inflows	−0.695601 (.830)	−4.374545* (.002)	−0.41478 (0.894)	−7.508179* (0)	Series: I(1)
Exports	1.550763 (.999)	−4.146115* (.003)	1.630998 (.999)	−4.160622* (.003)	Series: I(1)
Exchange rate	−2.458117 (.135)	−3.883315* (.005)	−2.169811 (.220)	−3.986590* (.004)	Series: I(1)

For ADF test, the values are at optimal lags, which are selected automatically by E-views using SIC.
For PP Test, the values are at Newey–West Bandwidth Criterion using Bartlett Kernel.
Figures in parentheses indicate *p* values.
*signifies statistically significant at 1% level.

Table A2. Unit root test with structural break.

Unit root testing with structural break					
	Zivot–Andrews unit root test (1992)		Perron unit root test (1997)		
Variables	t-stat.: I(0)	t-stat.: I(1)	t-stat.: I(0)	t-stat.: I(1)	Result
GDP	−2.894190	−5.322380*	−2.901367	−5.20447**	Series: I(1)
FDI inflows	−4.575828	−5.861549*	−4.508315	−6.016742*	Series: I(1)
Exports	−3.496080	−4.745760**	−3.426943	−4.588018**	Series: I(1)
Exchange rate	−3.429226	5.861549**	−3.668854	−4.945950**	Series: I(1)

The values are at optimal lags, which are selected automatically by E-views.
*signifies statistically significant at 1% level.
**signifies statistically significant at 5% level.

Table A3. Lag order selection.

VAR lag order selection criteria						
Lag	LogL	LR	FPE	AIC	SC	HQ
Endogenous variable: D(LGDP)						
0	78.15678	NA*	0.000471	−4.838399	−4.461214	−4.720269
1	80.24758	2.883865	0.000439*	−4.913626*	−4.489293*	−4.780730*
2	80.25191	0.005667	0.000474	−4.844959	−4.373478	−4.697297
Endogenous variable: D(LFDI)						
0	−15.26446	NA*	0.295596*	1.604445*	1.981630*	1.722575*
1	−14.64743	0.851063	0.305485	1.630858	2.055191	1.763753
2	−14.6468	0.000836	0.330011	1.699779	2.171260	1.847441
Endogenous variable: D(LEXPORT)						
0	40.85466	NA*	0.006164	−2.265839	−1.888654	−2.147709
1	42.15367	1.791741	0.006077*	−2.286460*	−1.862127*	−2.153564*
2	42.87190	0.941127	0.006248	−2.267028	−1.795546	−2.119366
Endogenous variable: D(LEXRATE)						
0	76.34658	NA*	0.000533	−4.713557	−4.336372	−4.595428
1	79.00373	3.665028	0.000479*	−4.827843*	−4.403510*	−4.694947*
2	79.55037	0.716288	0.000498	−4.796577	−4.325096	−4.648915

LR: sequential modified LR test statistics (each at 5% level), FPE: final prediction error; AIC: Akaike information criterion; SC: Schwarz information criterion; HQ: Hannan–Quinn information criterion.
*indicates lag order selected by the criterion.

Table A4. Diagnostic tests.

Test statistics	Dependent variable		
F-statistic	GDP ARDL(1,2,0,1)	FDI inflows ARDL (2,0,0,2)	Exports ARDL(1,1,0,2)
Serial correlation*			
F(1,21)	0.2411E − 3 (.988)	0.045555 (.833)	0.71867 (.406)
Heteroscedasticity**			
F(1,28)	0.9076E − 3 (.976)	1.0709 (.310)	1.7335 (.199)

*Lagrange multiplier test of residual serial correlation.
**Based on the regression of squared residuals on squared fitted values.
Figures in parentheses indicate *p*-values.
ARDL model selected based on Schwarz Bayesian Criterion.